

• Easy • #66 • Plus One •

▲ Problem Statement • Given a large integer represented as an array of digits, add "1" to the number & return the resulting array.

▲ Main Solution • Start from last digit because addition happens from right to left.

• If the digit is less than "9", simply add "1" & return.

• If the digit is "9", change it to "0" & carry "1" to the previous digit.

• If all digits are "9", create space for a new leading "1".

▲ Brute Force • Convert the array into a number, add "1", & convert it back into an array.

This is not suitable for very large integers because the number may exceed normal integer limits.

▲ Algorithm • 1] Start from the last digit.

2] If the current digit is less than "9" :- • increment by "1".

• Return the array.

3] If the digit is "9" :- Change it to "0".

• Continue to the previous digit.

4] If all digits become "0", insert "1" at the beginning.

5] Return the result.

▲ Important Points :- Start from the last digit because addition starts from the right.

• A "9" becomes "0" & creates a carry.

• If every digit is "9", add "1" at the beginning.

▲ Complexity - Brute Force → "Not Suitable for larger No." & $O(n)$
Main Solution → $O(n)$ & $O(1)$